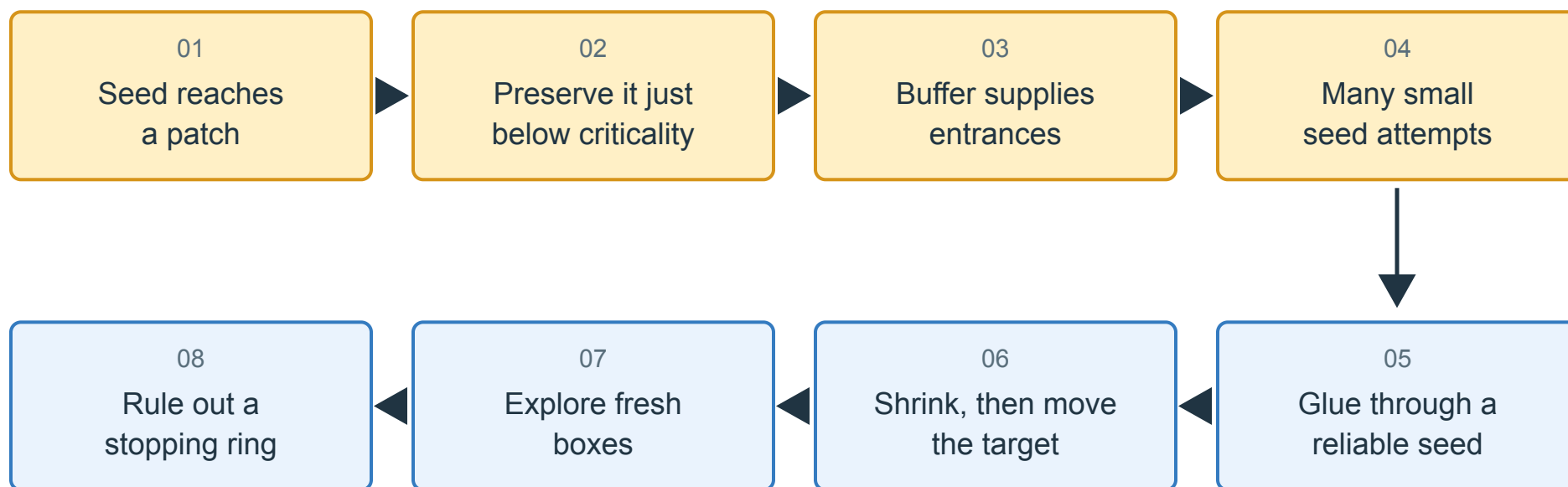


The proof, as a sequence of pictures

Assume an infinite cluster exists at criticality. Use that assumption to build an infinite route at a slightly lower density. That contradiction is the destination.



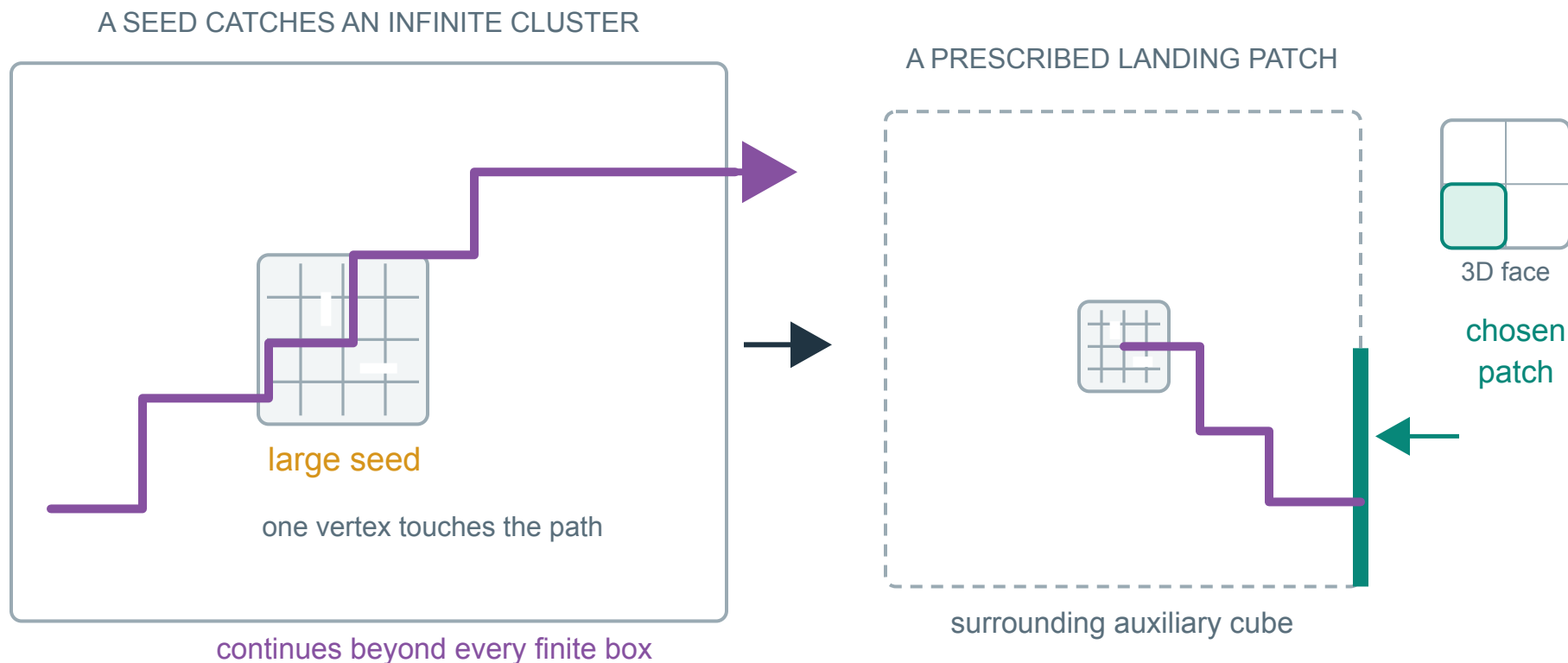
The growing route stays inside the original three-dimensional lattice.

We take the gluing theorem as available. The matrix and cone proof stay offstage.

The pictures simplify shapes and distances. Half-edges in flat drawings represent quarter-faces in three dimensions.

A large seed gives a reliable way out

A large seed is very likely to touch an infinite cluster under the initial assumption. That supplies a path to the boundary of any larger surrounding cube.

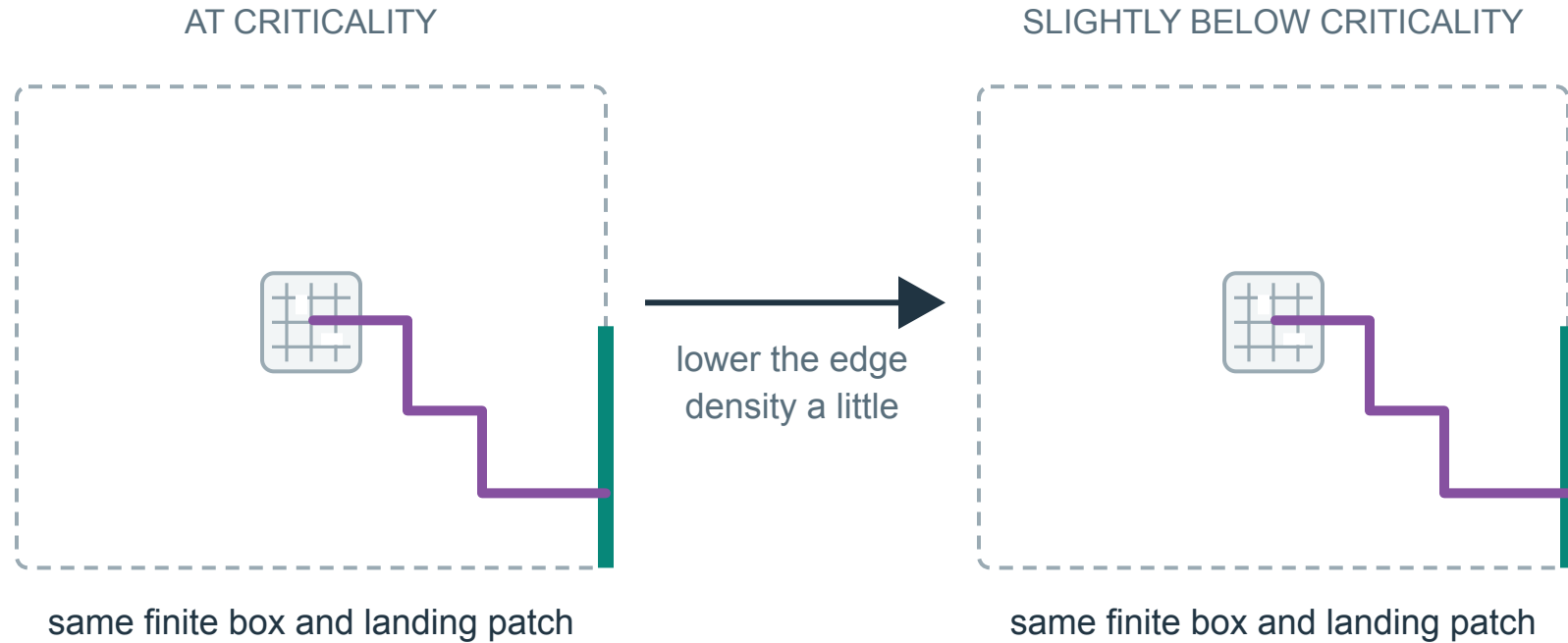


Symmetry and FKG upgrade “somewhere on the boundary” to “this prescribed quarter-face.”

The patch is chosen before seeing the configuration. The seed does not have to be fully open at this stage. The purple route is one illustrative successful configuration.

Carry the local crossing below criticality

Once all the finite boxes needed by the construction are fixed, lower the edge density slightly. Their crossing probabilities change only slightly, so the useful local crossings remain reliable.



Everything after this page runs at the lower density.

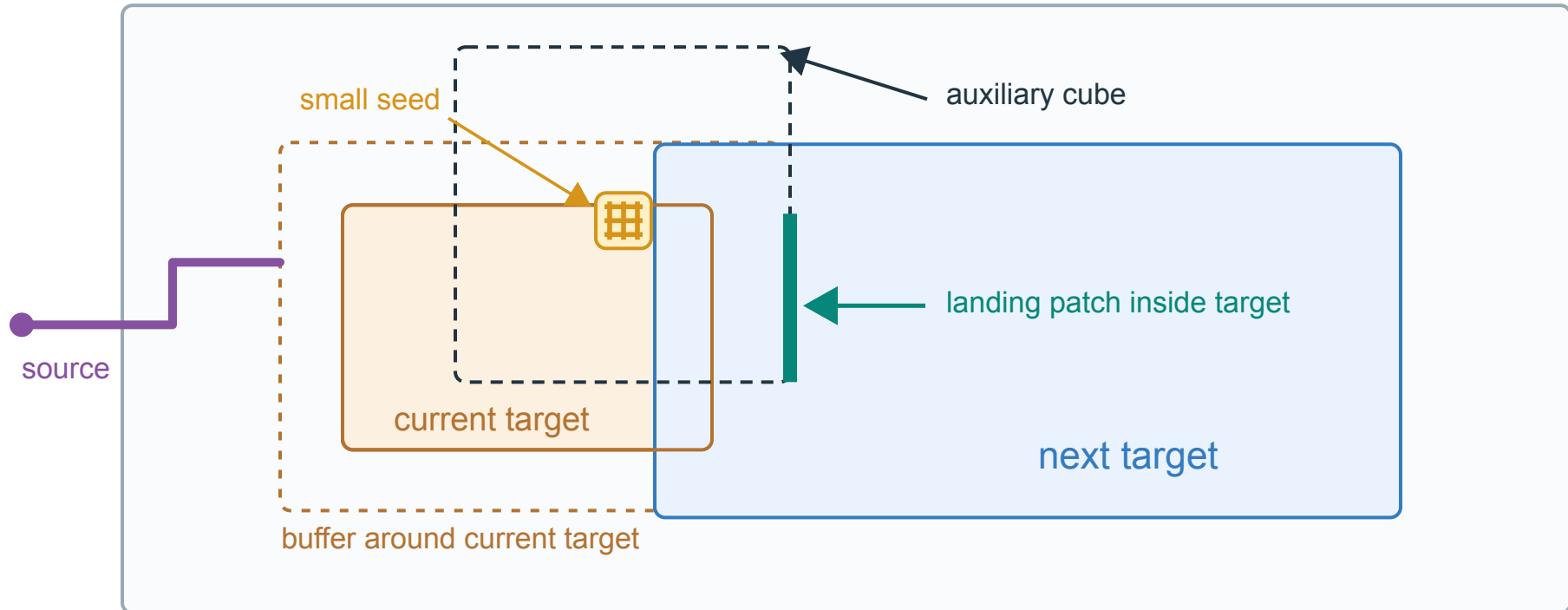
We preserve a finite collection of local crossing tests. Infinite growth is built later.

The matching routes are schematic: preserving a crossing probability does not mean preserving the same realized path.

One extension has several different regions

The goal is to make “source reaches the current target but misses the next one” a rare outcome. The buffer supplies entrances; small seeds attach paths; auxiliary cubes supply onward crossings.

FRESH REGION: hidden edges still have their original independent law



The next target may overlap the current target. Only the landing patch must fit inside it.

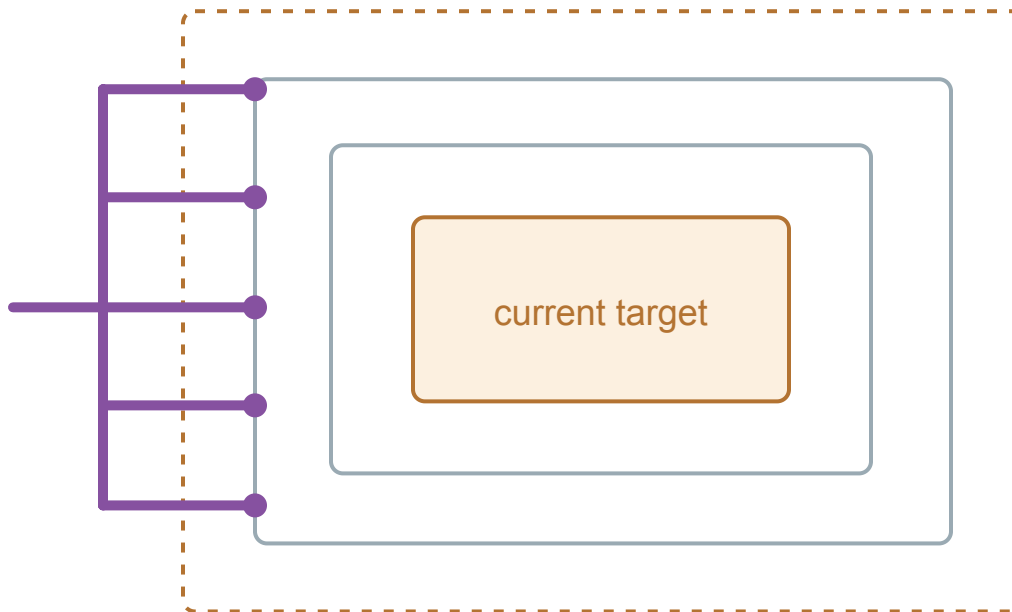
The whole auxiliary cube stays in fresh space. Its selected quarter-face lands in the next target.

The current target is a fixed set. We are estimating a joint event; we have not first revealed and conditioned on a successful connection to it.

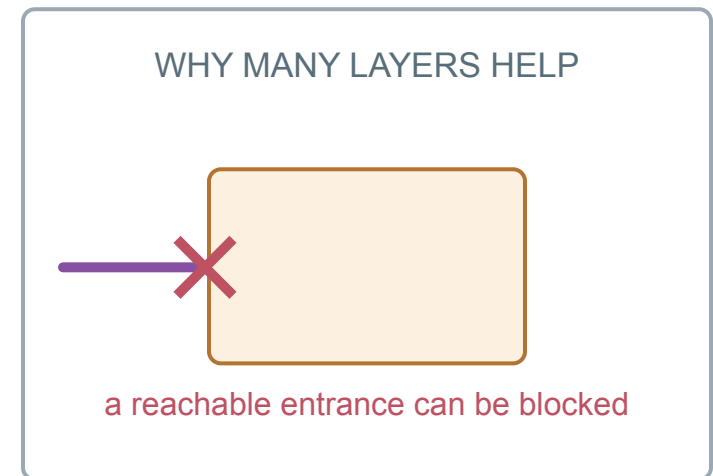
The buffer supplies many reachable entrances

Imagine nested fences. If only a few gates are reachable, closing those gates can block the source. Such reachable, completely blocking fences cannot occur at several nested layers at once.

incoming cluster



buffer: many candidate surrounding layers



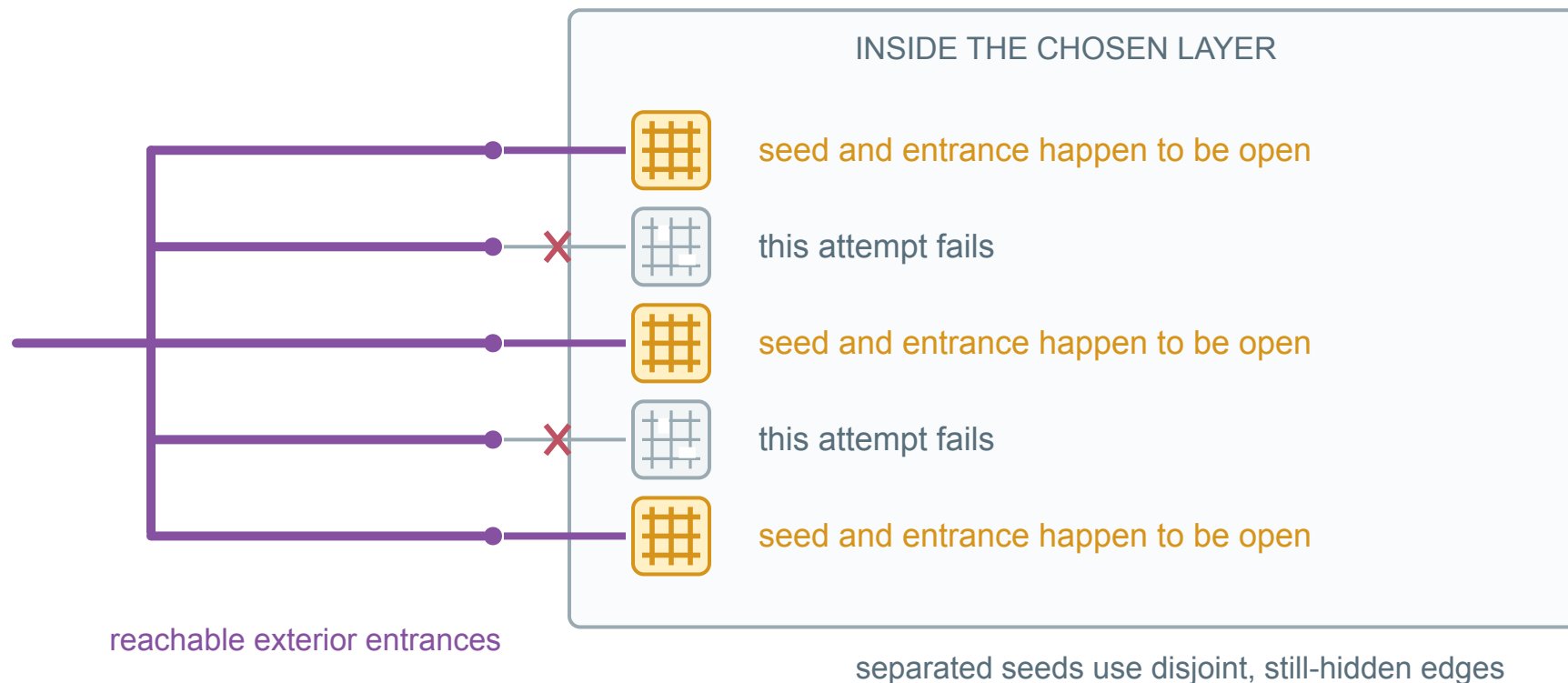
A blocked outer layer prevents access to every smaller inner layer.

This finds a fixed layer where reaching the current target with only a few entrances is rare.

The layer is chosen in advance from probability estimates. We do not inspect a configuration and then pick its luckiest layer.

Many small trials pay for fully open seeds

A fully open seed is rare. We keep the seed size fixed and create many separated opportunities. In a successful trial, the entrance edge and every internal seed edge are open.

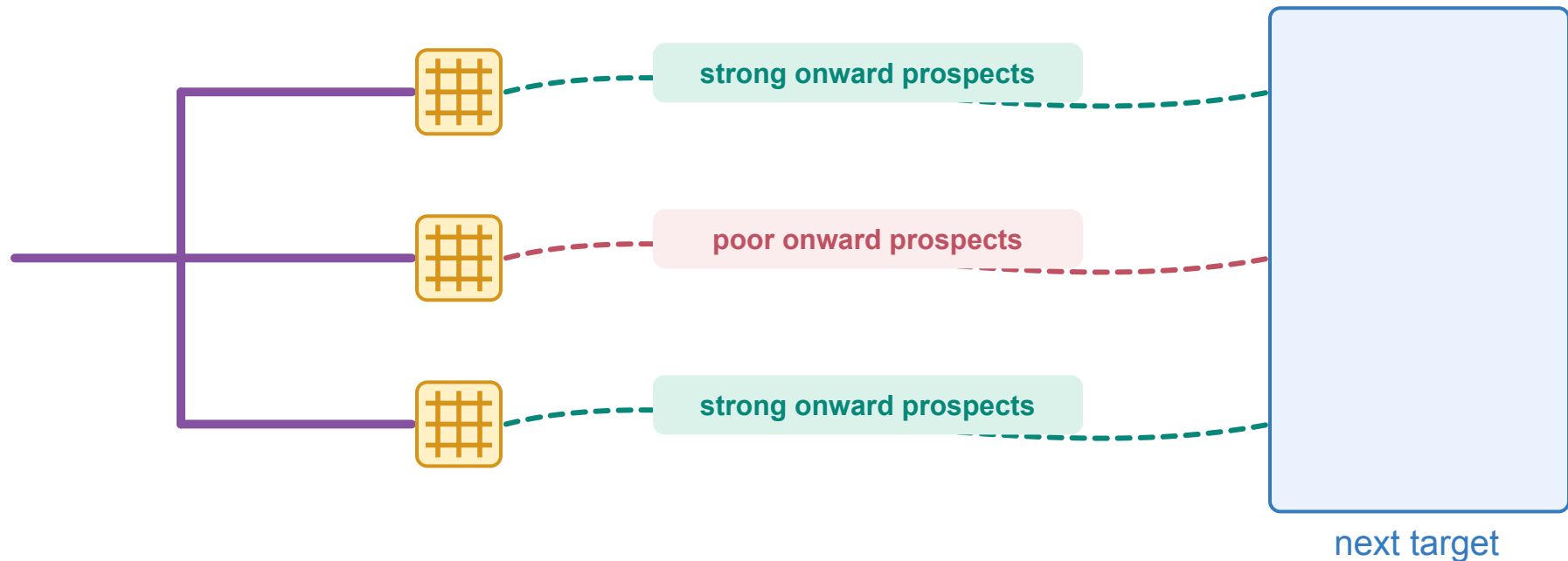


Full opening joins the incoming path to every vertex in that seed, including any vertex with an onward route.

We look for naturally open seeds in one random configuration. We do not force edges open or reroll them. These sample successes are illustrative.

Reach a reliable seed, then use gluing

Most fully open seeds have strong onward prospects after the interior is fixed. Many successful trials leave a reached seed with that property. Gluing then controls failure to continue to the target.



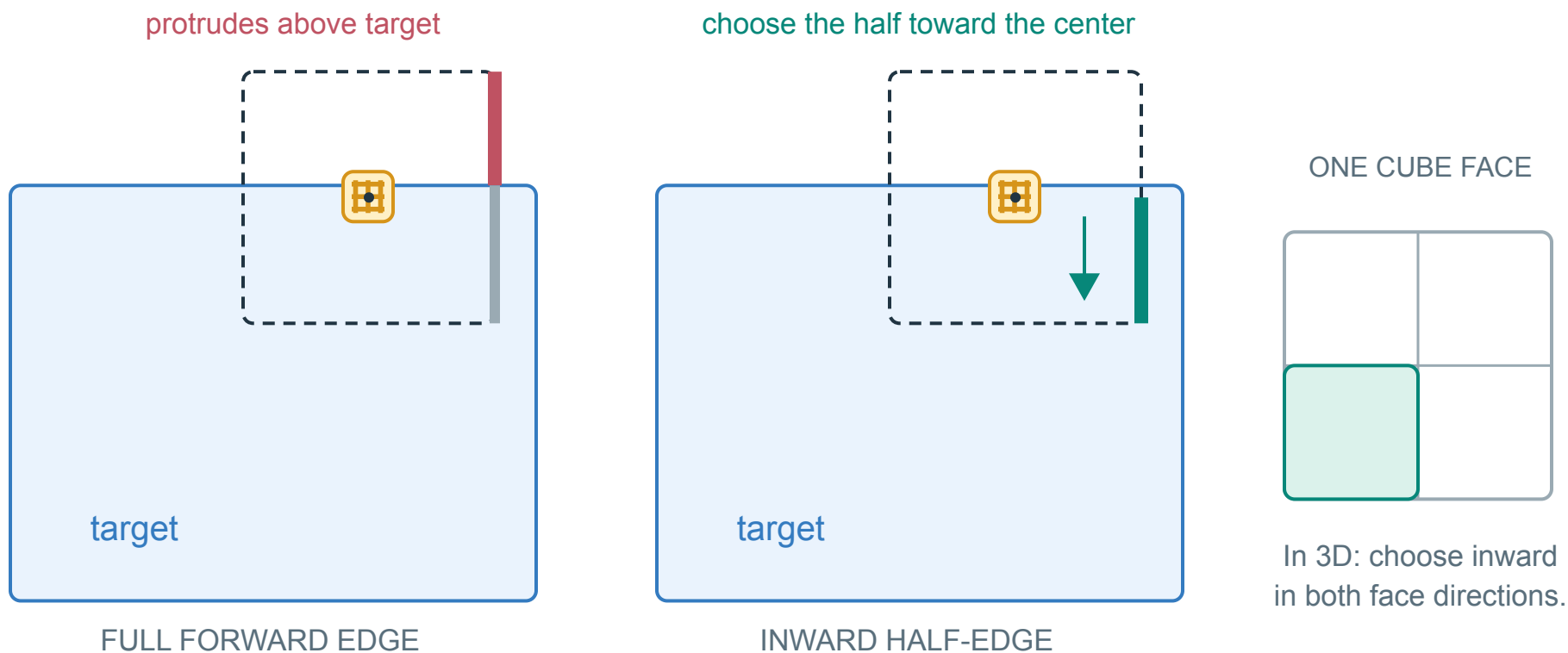
Gluing joins “reach some reliable seed” to “continue toward the target.”

We do not need to select the first reached seed and pretend its future is unbiased. Gluing handles the dependence.

Dashed curves show conditional connection prospects, not already revealed open paths. Seed-quality labels can be dependent.

Aim the landing patch inward

A seed can sit near an edge or corner of the allowed region. Its auxiliary cube's full forward face may stick out of the next target. Choosing the inward half in each sideways direction gives a quarter-face that fits.



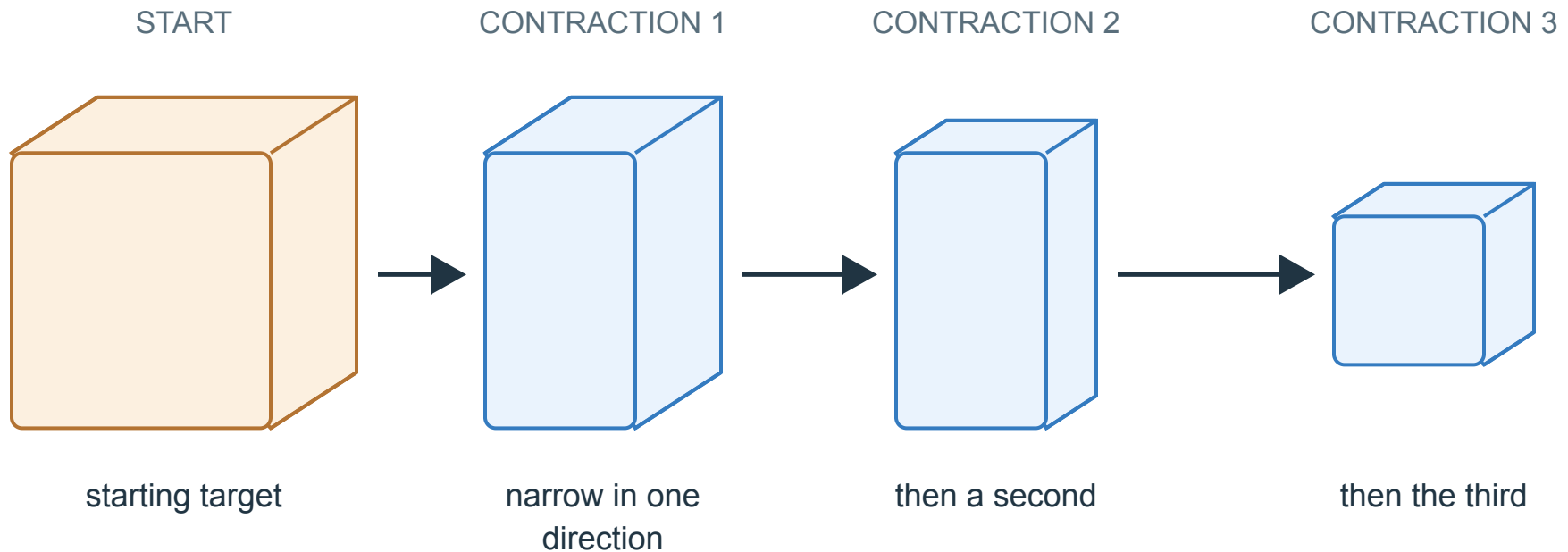
The auxiliary cube can stick out of the target; the selected landing patch cannot.

The forward position is checked separately. The inward choices keep the patch inside the target sideways.

Flat drawings use a half-edge. The inset shows the actual quarter of a square face in three dimensions. Fresh space around the auxiliary cubes is omitted here.

Three contractions create room to move

Before crossing to the neighboring coarse box, use three extension steps to aim for a narrower target. That leaves spare width for the buffer allowances accumulated during the later moves.



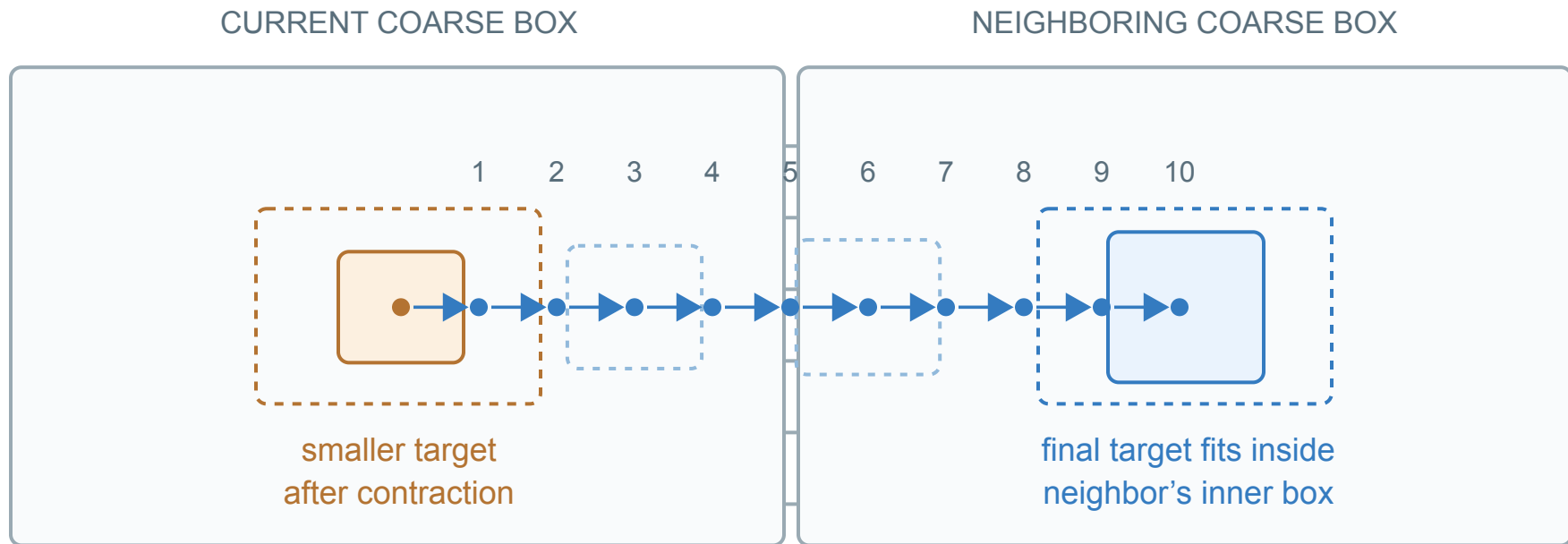
The target shrinks; the actual cluster is not squeezed.

These are successive target sets in the same configuration. Their centers stay at the same place during contraction.

The shapes are separated for comparison. Small buffer enlargements are suppressed so you can see the principal contraction in each direction.

Ten translations reach the neighboring box

Move the target toward the neighboring box in ten short steps. It grows a little at each step to allow for the next buffer. The initial contraction leaves enough space for the final target to fit.



TEN TRANSLATIONS

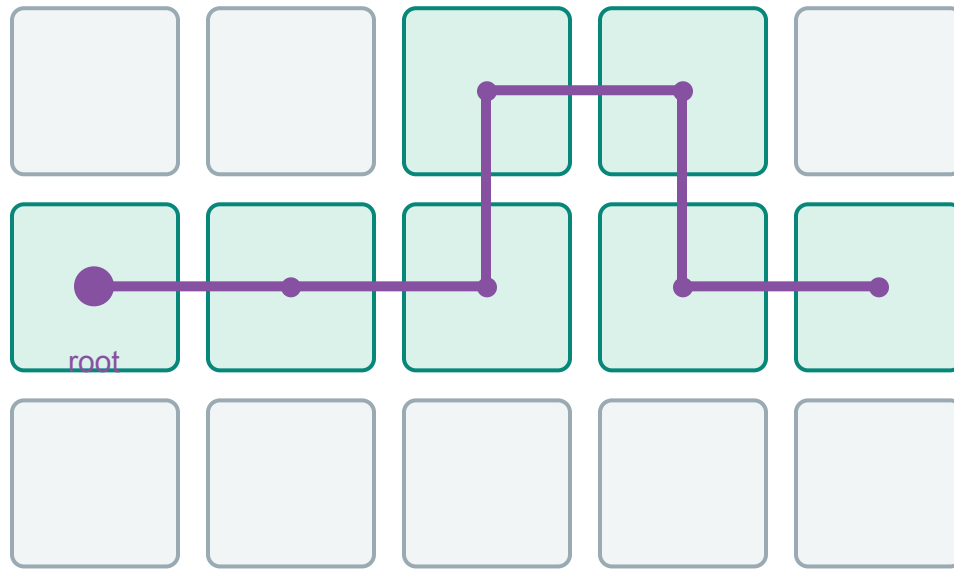
Each move uses the same buffer → seeds → quarter-face → gluing extension.

The result: reaching this box's inner target and missing its neighbor's inner target is unlikely.

Thirteen is the number of moves in this explicit construction. The argument does not assume that their success events are independent.

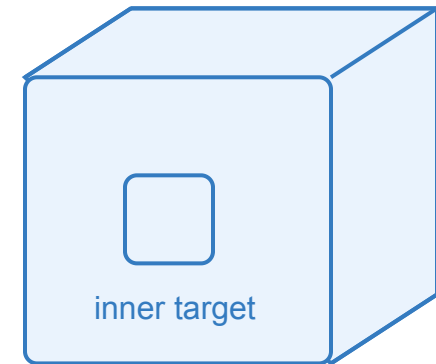
Repeat the relay on a planar array of thick boxes

Place the coarse boxes in a square grid, with their centers in one plane. Each box still contains a full three-dimensional region. An infinite connected route through this array is already enough.



top view: a two-dimensional array of boxes

EACH BOX HAS THICKNESS



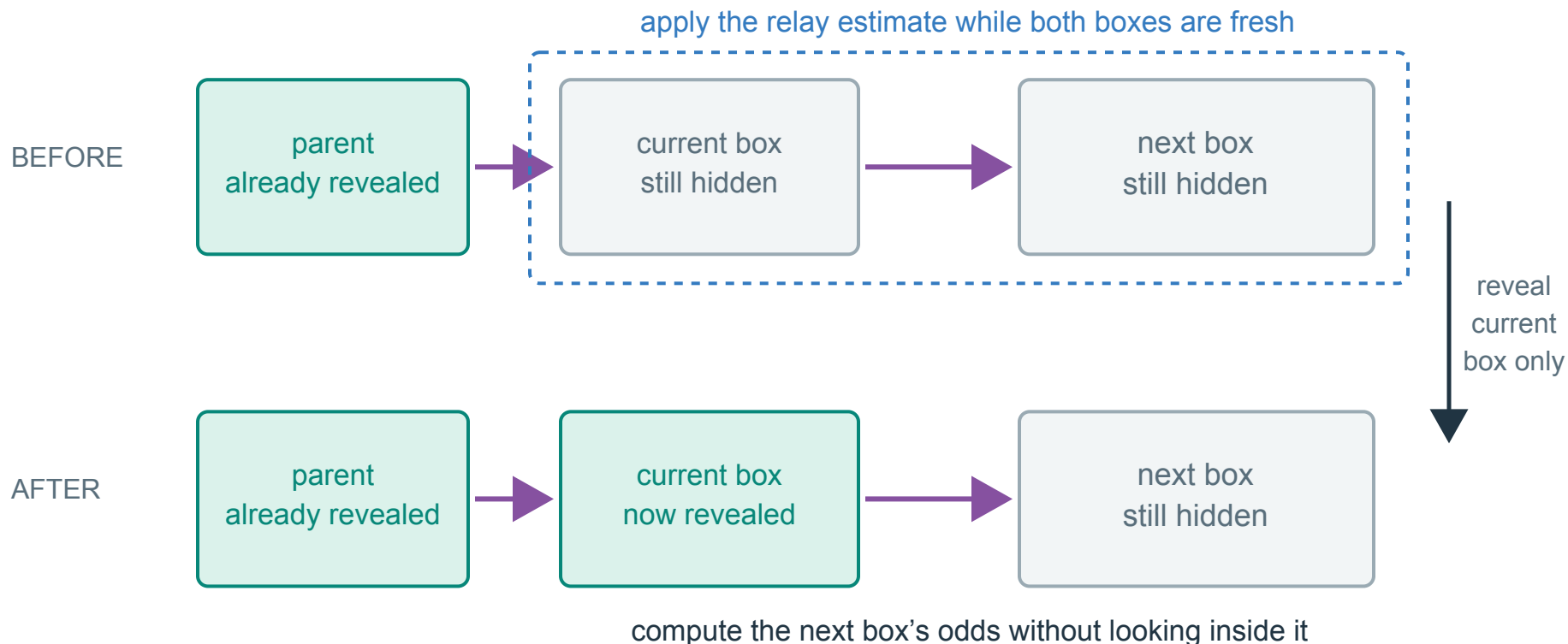
Paths can move in all three directions inside every box.

The exploration starts on a positive-probability event where the root box and its outgoing interfaces are open.

That finite initial cost is paid once. Every later box must pass a reliable exploration test. The planar arrangement will make the final stopping boundaries easy to count.

Protect tomorrow's edges while exploring today

A pending box comes with a strong prediction of connection through its parent chain. Until its turn, nobody else reveals its interior or its incoming parent interface. Its prediction therefore stays valid.



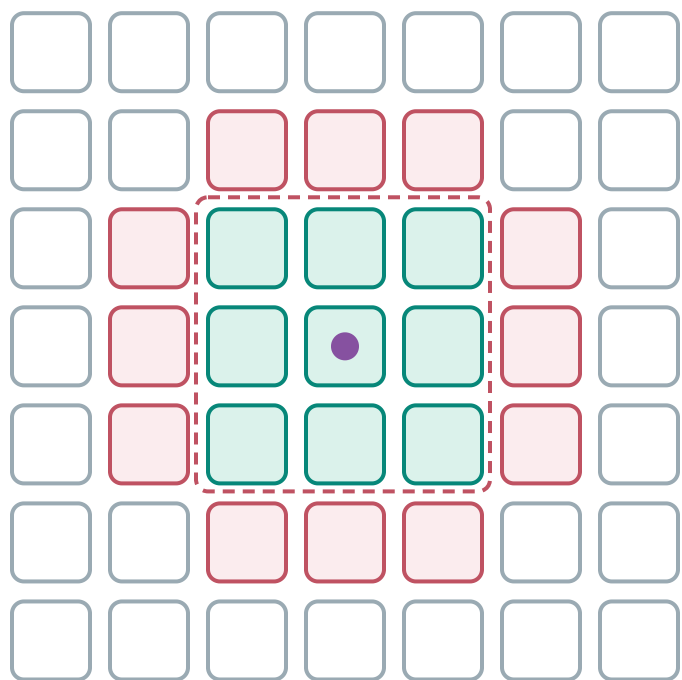
Fresh means the hidden edges retain their original independent law, given the entire revealed history.

A box is declared good only after its own connection is verified and its unvisited neighbors have strong connection predictions. Bad boxes produce no children.

A stopped exploration would leave a bad boundary

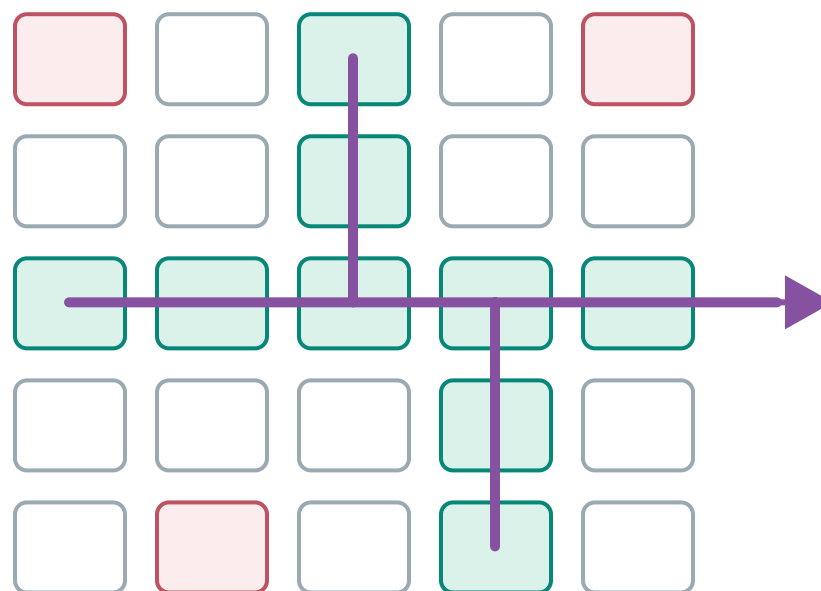
When the exploration finishes, every outside neighbor of its finite good cluster has been processed and is bad. In the planar grid, the boundary contains a surrounding dual loop.

IF GROWTH STOPS



a surrounding dual loop crosses toward bad neighbors

SOME GROWTH ESCAPES



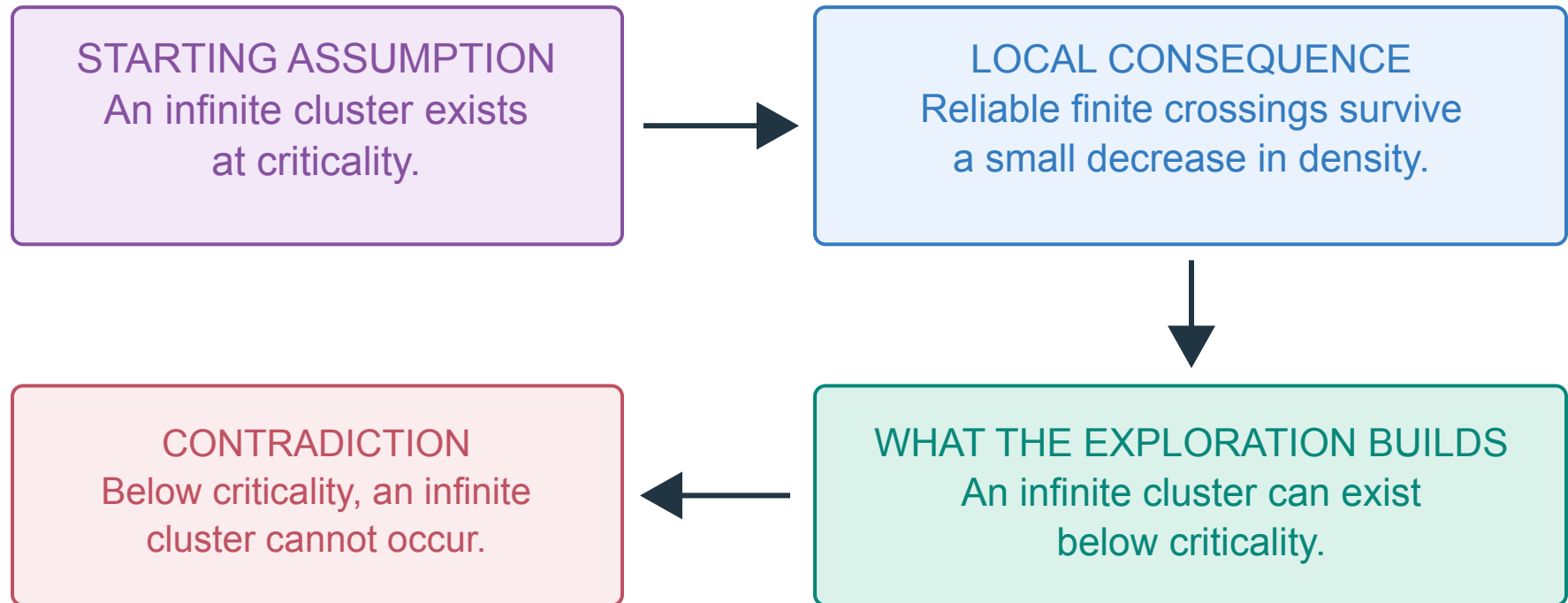
an unending route has positive probability

Counting loops and controlling bad boxes leaves a positive chance that the exploration never stops.

Bad-box decisions need not be independent. Section 9 supplies conditional bounds strong enough to control any specified collection of processed bad boxes.

The route cannot exist below criticality

Every good box has a verified connection back to the original source. Infinitely many good boxes therefore put infinitely many vertices in that source's actual open cluster.



The starting assumption must fail: the critical configuration has no infinite open cluster.

Seed crossings supply the local opportunity. Gluing makes it usable. Geometry repeats it. Protected exploration and planar contours turn it into infinite growth.